

Markov vs Martingale Processes

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Markov Process

A Markov process (or Markov chain in discrete time) is a stochastic process where the future state depends only on the present state and not on the past states. This property is known as the Markov property.

Mathematically: If X_t is a stochastic process, it is a Markov process if:

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} \mid X_t)$$

This means that the probability of moving to the next state X_{t+1} depends only on the current state X_t and not on any previous states $X_{t-1}, X_{t-2}, \dots$.

Examples:

- ▶ Stock price movements (under some assumptions)
- ▶ Weather modeling (if today is rainy, the probability of tomorrow being sunny depends only on today)
- ▶ Queueing systems (e.g., customers arriving at a bank)

Martingale Process

A Martingale is a stochastic process where the expected future value is equal to the present value, given all past information. This means that there is no predictable trend in the process.

Mathematically: A process X_t is a Martingale if:

$$E[X_{t+1} \mid X_t, X_{t-1}, \dots, X_0] = X_t$$

This means that given the present state, the expected value of the next state remains the same.

Key Properties:

- ▶ **No Drifting Trend:** A Martingale does not have an upward or downward bias.
- ▶ **Fair Game:** In finance, this represents a "fair game" where future expected gains or losses are zero.

Examples:

- ▶ **Gambling:** A fair coin toss betting system where the expected earnings at the next round are equal to current earnings.
- ▶ **Stock Prices in an Efficient Market:** If markets are truly efficient (following the Efficient Market Hypothesis), stock prices follow a Martingale process because their expected future value is the same as the current price.

Markov vs Martingale Processes:

- ▶ A Martingale does not necessarily have the Markov property because its next value may depend on the entire history.
- ▶ However, some Martingale processes can also be Markov processes if their future states depend only on the current state.